DEVELOPMENT, NON-LINEAR MODELING, AND EMBEDDED TACTILE CONTROL OF A 3D-PRINTED COMPLIANT SOFT ROBOTIC GRIPPER

BTP-I Project Stage Presentation | Mechanical Engineering (Manufacturing Science & Engineering)

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Introduction to Soft Robotics & Compliant Manipulation

Overcoming the limitations of traditional rigid robotic end-effectors

Conventional Rigid Grippers

- Constructed from metallic links and discrete kinematic joints.
- High positioning repeatability and rigidity in structured environments.
- Point contact creates localized stress spikes on fragile workpieces.
- Significant risk of crushing biological or delicate objects.
- Requires complex multi-axis trajectory planning and force sensing.

Soft & Compliant Grippers

- Replaces rigid links with continuously deformable elastomeric bodies.
- Embodies mechanical intelligence directly into structural compliance.
- Passively conforms to unknown, irregular 3D object geometries.
- Distributes interaction forces over an expanded contact area.
- Inherently safe for agricultural produce, medical vials, and humans.

Background and Industrial Motivation

Target applications demanding gentle, adaptive, and sensorized manipulation

Automated Agriculture

- Selective harvesting of soft-skinned fruits and vegetables (tomatoes, strawberries).
- Eliminates internal tissue bruising that accelerates biological decay.
- Adapts to natural variations in fruit size and curvature.

Biomedical Logistics

- Handling thin-walled glass vials, filled fluid ampoules, and test tubes.
- Prevents mechanical shock and brittle fracture during rapid sorting.
- Conforms gently to anatomical specimens and surgical tools.

Explosive Ordnance Disposal

- Interfacing with uncertain, aged, or structurally compromised munitions.
- Spark-free polymer construction eliminates electrical discharge.
- Prevents impact shock that could trigger detonation.

Formal Problem Statement & Key Challenges

Bridging the gap between structural compliance, payload capacity, and tactile perception

"Conventional rigid end-effectors lack the compliance required to manipulate fragile objects without damage, while existing soft grippers suffer from low payload capacity, material hysteresis, tethered fluidic supplies, and an absence of localized feedback capable of detecting incipient slip without bulky optical systems."

Critical Engineering Bottlenecks

- Non-linear hyperelastic material behavior requiring specialized FEA.
- Viscoelastic relaxation during sustained holding tasks.
- Planar sensor delamination and peeling on stretching soft skins.

Proposed Engineering Solution

- Monolithic 3D-printed TPU 95A structure featuring circular living hinges.
- Tendon-driven actuation with remote base-mounted servomotors.
- Flush-recessed FSR402 tactile sensors + on-chip FFT slip detection.

Consolidated Literature Review: Team Synthesis

Integration of individual research streams into a unified knowledge base

Actuation & Kinematics (Sanchali)

- Evaluated pneumatic, tendon, SMA, and DEA architectures.
- Mobility classification: constrained, underconstrained, deformable.
- Highlighted payload vs compliance trade-off in fluidic bladders.
- Base motors reduce distal finger inertia.

Materials & Fabrication (Chandana)

- Evaluated silicones (Ecoflex, PDMS), TPUs, TPEs, and photopolymers.
- Molding requires multi-step casting, degassing, and curing ovens.
- Selected FDM 3D printing of TPU as the optimal fabrication route.
- Eliminates tooling, enables rapid prototyping.

Tactile Transduction (Saurav)

- Surveyed piezoresistive, capacitive, piezoelectric, and optical sensors.
- Decoupled static contact force from dynamic slip micro-vibrations.
- FSR402 thin-film piezoresistors provide ideal form factor.
- Requires active analog filtering & drift mitigation.

Comparative Analysis of Actuation Paradigms

Multi-attribute engineering trade-offs governing methodology selection

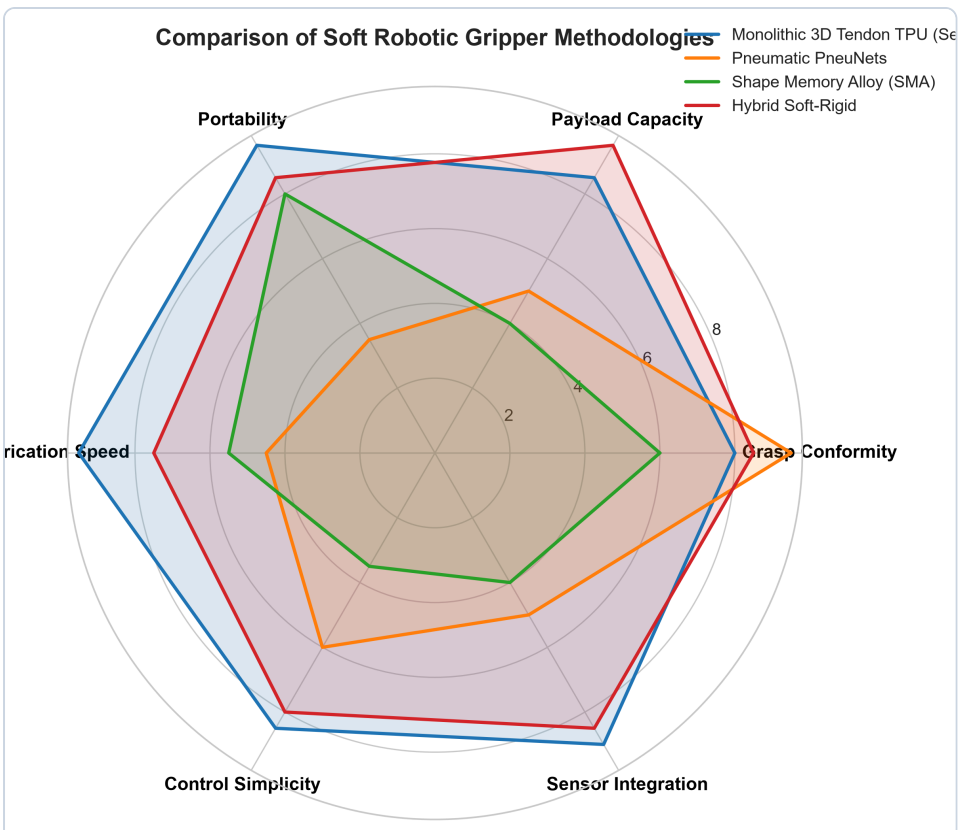


Figure: Multi-attribute radar comparison across 6 engineering dimensions.

Comparative Engineering Takeaways

- **Pneumatic PneuNets:** Maximum passive conformity, but tethered compressor, bulky valves, and seam rupture risks under pressure.
- **Shape Memory Alloy (SMA):** High energy density, but slow cooling lag (3–15 s), severe thermal hysteresis, and substrate degradation.
- **Dielectric Elastomers (DEAs):** Fast response, but hazardous driving voltages (>1–5 kV) and breakdown risks.
- **Monolithic Tendon TPU (Selected):** High payload ratio (3:1 to 5:1), completely untethered battery operation, rapid FDM printing, and straightforward servomotor PWM control.

Identified Research Gaps in Published Literature

Explicit limitations motivating our proposed methodology

Gap 1: Compliance vs. Payload Capacity

- Pure elastomeric bladders buckle under lateral/torsional loads.
- Payloads typically restricted to <200–400 grams.
- Need for flexure mechanisms combining compliance with structural load capacity.

Gap 2: Interfacial Sensor Delamination

- Planar sensors bonded to stretching silicone skin experience severe peeling shear.
- Cyclic bending causes electrical trace rupture and sensor detachment.
- Need for recessed, strain-isolated sensor pockets on rigid contact pads.

Gap 3: Bulky Slip Detection Systems

- State-of-the-art slip sensing relies on heavy optical cameras (GelSight).
- Computationally intensive image processing unsuited for mobile arms.
- Need for lightweight, edge-computed tactile vibration analysis.

Gap 4: Embedded Edge Control Disconnect

- Many research prototypes rely on desktop MATLAB scripts.
- Lack of deterministic real-time operating systems running on-board.
- Need for FreeRTOS dual-core embedded control coupled to ROS 2.

Project Objectives & Technical Scope

Specific measurable targets across mechanical, sensing, and control domains

Main & Specific Sub-Objectives

- **Main:** Establish the design, modeling, fabrication, and control methodology for a compliant gripper.
- **Design Strategy:** Define circular living hinge geometry using Howell PRBM formulations.
- **Material Selection:** Formulate direct-drive FDM printing with flexible TPU 95A filament.
- **Simulation Strategy:** Formulate hyperelastic constitutive modeling for BTP-II simulation.
- **Signal Conditioning:** Design Sallen-Key 2nd-order Butterworth LPF ($f_c = 20extHz$).
- **Firmware Architecture:** Design dual-core FreeRTOS pipeline (1 kHz FFT / Servo PWM).
- **Validation Plan:** Outline pull-out force benchmarking and autonomous slip prevention.

Technical Scope & Boundaries

- **COMPLETED IN STAGE-I:**
- • Multi-thematic literature review & comparative technology evaluation.
- • Selection of monolithic FDM TPU tendon-driven gripper methodology.
- • Analytical kinematic formulations (Howell PRBM & Capstan friction).
- • Embedded signal processing & FreeRTOS architectural design.
- **PLANNED FOR STAGE-II (BTP-II):**
- • SolidWorks 3D CAD modeling & Ansys non-linear hyperelastic FEA.
- • Physical prototype 3D printing, sensor assembly, & experimental trials.
- **OUT-OF-SCOPE:**
- • High-speed sorting (>3 m/s), multi-axis shear arrays, underwater trials.

Confirmed Methodology: Monolithic 3D-Printed Tendon Gripper

Synthesis of compliant mechanisms, additive manufacturing, and embedded sensing

1. Structural Compliance

- Monolithic FDM 3D printing with flexible TPU 95A.
- Replaces discrete mechanical pins with living hinges.
- Zero joint assembly, zero backlash, zero lubrication.
- Horizontal build orientation maximizes inter-layer tensile strength.

2. Tendon Transmission

- Remote base-mounted metal-gear digital servomotors (DS3218).
- Braided Dyneema cable (>30 kg rating) pulling through low-friction guides.
- Micro-PTFE tubing inserts minimize Capstan friction losses.
- Passive elastic restoration driven by stored TPU strain energy.

3. Embedded Intelligence

- FSR402 thin-film sensors recessed flush in contact pads.
- Analog 2nd-order Butterworth filtering for normal force F_n.
- On-chip 256-point FFT on ESP32 Core 0 for slip vibrations.
- Closed-loop servomotor torque tightening in <35 milliseconds.

Kinematic Modeling & Pseudo-Rigid-Body Formulation

Analytical PRBM living hinge stiffness, friction modeling, and underactuated contact mechanics

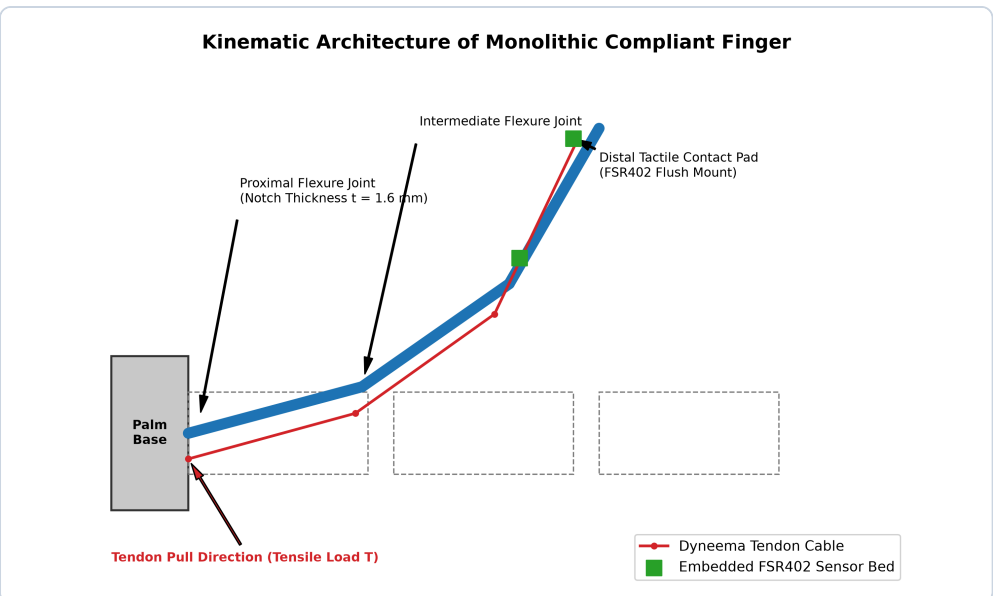


Figure: Kinematic profile of the monolithic compliant finger under cable tension.

Compliant Living Hinge Mechanics (PRBM)

- **Howell PRBM Formulation:** Circular flexure notches modeled as torsional spring joints: **\(K_ heta = \frac{E \cdot w \cdot t^{2.5}}{9 \pi \sqrt{r}}\)** $(t = 1.6 extmm, r = 2.5 extmm, w = 20 extmm)$
- **Capstan Friction Law:** $T_{extdistal} = T_{extbase} \cdot e^{-\sum \mu_f heta_i}$
- Micro-PTFE sleeves minimize friction to $\mu_f \leq 0.08$, maximizing force delivery.

Underactuated Mechanical Intelligence & Grasping Sequence

- **1 Actuator → 3 DOFs:** Single Dyneema tendon drives MCP, PIP, DIP joints without complex multi-joint inverse kinematics.
- **Phase 1 (In-Flight Flexure):** Cable tension T flexes all 3 joints simultaneously proportional to compliance $K_{heta,i}$.
- **Phase 2 (Proximal Arrest):** Workpiece contact creates counter-moment $M_{extcontact}$, arresting proximal link rotation.
- **Phase 3 (Passive Conformal Curling):** Tension bypasses the arrested link through the PTFE sleeve, curling distal phalanges around object contours.
- **Phase 4 & 5 (Sensor Lock & Return):** Distal pad engages FSR402; slacking tendon releases stored elastic strain energy (\(E_{ ext{strain}} = \sum \frac{1}{2} K_{ heta, i} heta_i^2\)) for passive spring return.

Proposed Material Selection & Planned Simulation Strategy

Structural selection of TPU 95A and non-linear hyperelastic modeling approach for BTP-II

Material Selection: TPU 95A

- Evaluated silicones (Ecoflex, PDMS), photopolymers, and polyurethanes.
- Silicone molding requires multi-part molds, vacuum degassing, and curing ovens.
- Direct FDM 3D printing of TPU 95A eliminates molds and enables rapid prototyping.
- Shore 95A provides high flexural resilience while preventing out-of-plane torsional buckling.
- Monolithic printing enables continuous living hinges without assembly pins or mechanical backlash.

Planned Hyperelastic FEA Strategy (BTP-II)

- **Non-linear Mechanics:** TPU undergoes large elastic deformations and strain-stiffening. Hookean linear elasticity is invalid for large flexure bending.
- **Selected Constitutive Law:** 2-Parameter Mooney-Rivlin strain energy formulation:
  \(W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) + ↑rac{1}{D_1}(J - 1)^2\)
- **Planned BTP-II Simulation Goals:**
- • Model finger curl angle versus tendon tension prior to 3D printing.
- • Evaluate Von Mises stress concentrations at flexure notches to ensure high fatigue life.
- *Status: Modeling methodology decided; simulation runs will be performed in BTP-II.*

Mechatronic & Embedded Firmware Architecture

Hardware-software co-design uniting analog filtering, FreeRTOS, and micro-ROS

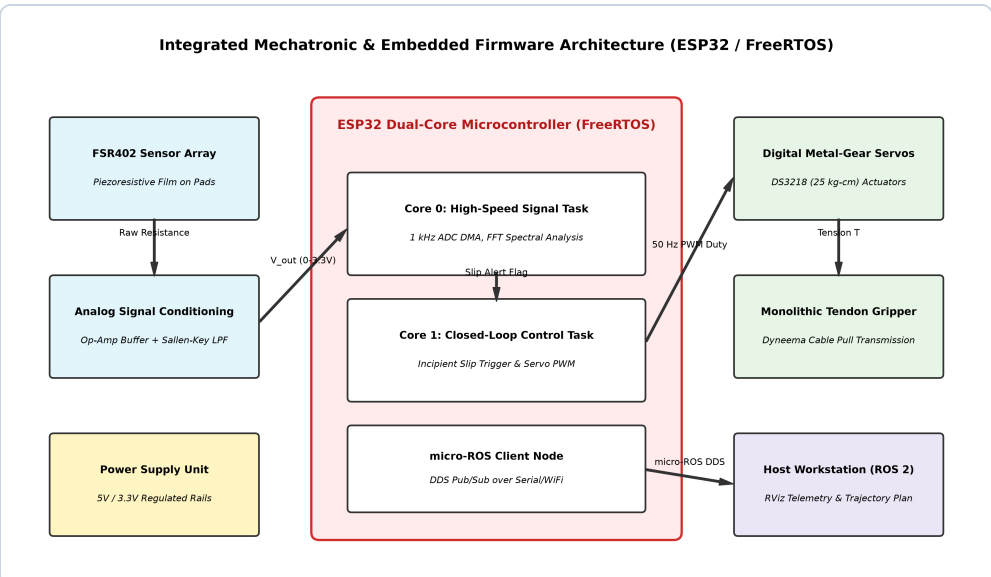


Figure: Integrated mechatronic and FreeRTOS embedded architecture.

Dual-Core FreeRTOS Pipeline

- **Core 0 (1 kHz DMA Task):**
  - High-speed 12-bit ADC sampling.
  - 2nd-order Butterworth LPF ($f_c = 20extHz$).
  - 256-point Radix-4 FFT spectral calculation.
  - Incipient slip threshold evaluation.
- **Core 1 (100 Hz Control Task):**
  - Closed-loop servomotor PWM torque update.
  - Hysteresis compensation lookup.
  - micro-ROS DDS publishing over serial/WiFi.

Tactile Signal Processing & Autonomous Slip Prevention

Real-time frequency-domain FFT vibration spike detection

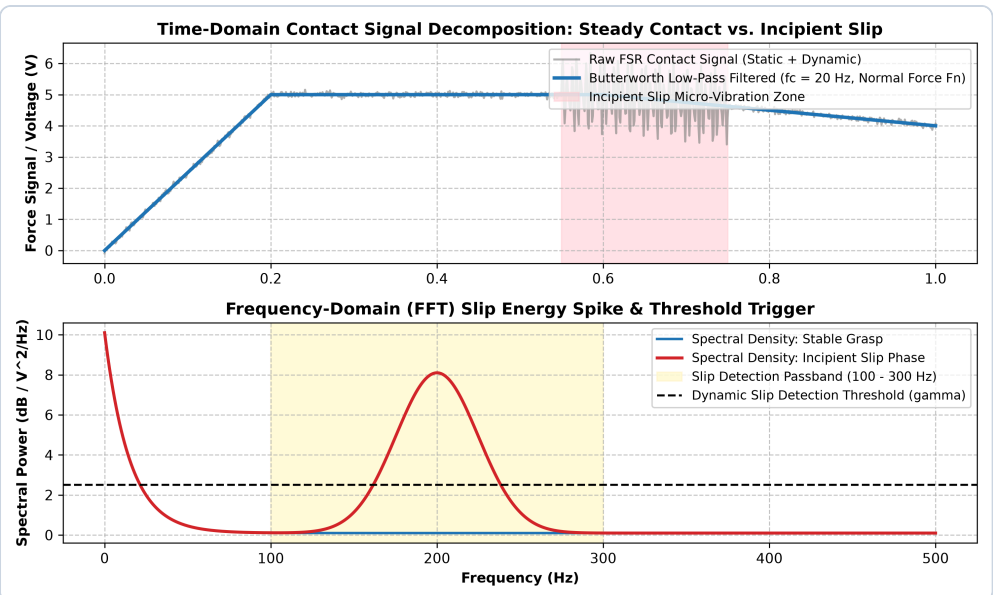


Figure: Tactile signal decomposition and FFT slip-detection pipeline.

Spectral Energy Triggering

- **Time Domain:** Normal clamping force F_n resides in low-frequency band (0–20 Hz).
- **Dynamic Friction:** Incipient micro-vibrations generate sharp tactile energy spikes in 100–300 Hz band.
- **On-Chip FFT Metric:** $E_{extslip} = \sum_{k=25}^{76} |X[k]|^2$
- **Autonomous Compensation:** When $E_{extslip}$ exceeds dynamic baseline threshold, ESP32 commands immediate servomotor torque boost in <35 ms, arresting slip.

Current Progress & Project Status (BTP-I Baseline)

Clear differentiation between completed conceptual decisions and proposed BTP-II execution

Work Completed in Stage-I

- • **Literature Review:** Comprehensive synthesis of actuation, materials, and tactile sensing.
- • **Methodology Decision:** Multi-criteria selection of Monolithic FDM TPU Tendon Gripper.
- • **Conceptual Design:** Selected circular living hinge architecture and Dyneema routing.
- • **Theoretical Kinematics:** Howell PRBM living hinge stiffness & Capstan friction equations.
- • **Signal Architecture:** Sallen-Key Butterworth filter & dual-core FreeRTOS pipeline.
- • **Implementation Plan:** Structured 16-week execution roadmap for BTP-II.

Status Clarification: Decisions & Planning

- **IMPORTANT ACADEMIC CLARIFICATION:**
- ***"We have only decided what we will do."***
- • In strict compliance with academic standards, no SolidWorks CAD models were constructed in Stage-I.
- • No Ansys FEA simulations were executed in Stage-I.
- • No physical prototypes were 3D printed or experimentally tested in Stage-I.
- • All 3D CAD modeling, FEA simulation, fabrication, sensor wiring, and experimental trials are scheduled for Stage-II (BTP-II).

Anticipated Engineering Challenges & Mitigations

Risk mitigation strategy formulated for next-semester manufacturing & testing

TPU Extrusion Jamming

- **Cause:** Flexible filament buckling in direct-drive gear under backpressure.
- **Mitigation:** Use direct-drive extruder with constrained path; print at 20 mm/s; disable retraction.

Inter-Layer Shear Cleavage

- **Cause:** Anisotropic Z-axis weakness during high flexural bending.
- **Mitigation:** Print fingers flat horizontally; set nozzle to 225°C to achieve 100% polymer weld.

Tendon Friction Hysteresis

- **Cause:** Dyneema cable abrading 3D-printed layer ridges in channels.
- **Mitigation:** Insert micro-PTFE tubing (OD 2 mm, ID 1 mm) along internal cable channels.

FSR Sensor Resistance Drift

- **Cause:** Conductive polymer relaxation under continuous load.
- **Mitigation:** Firmware implements dynamic AC baseline tracking, decoupling static and AC slip signals.

BTP-II Semester-Wise Implementation Roadmap

16-week timeline, task breakdown, and technical milestones

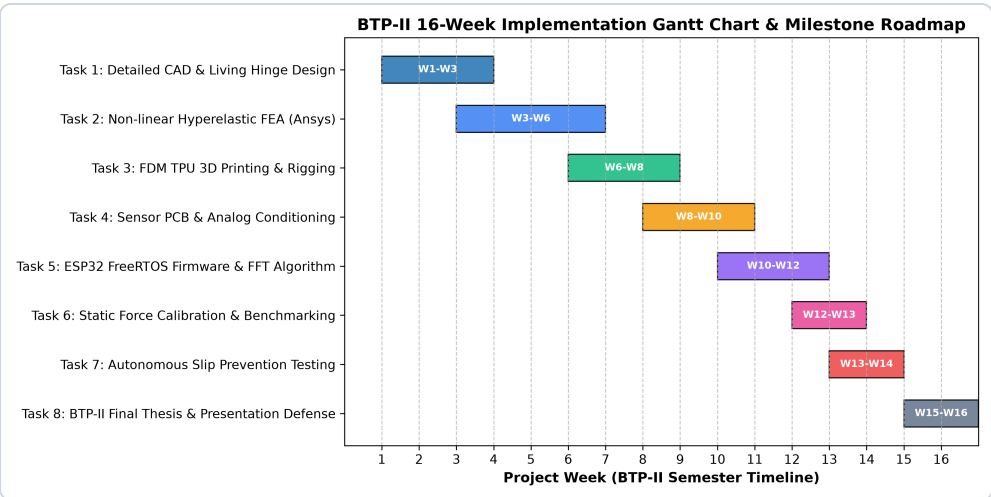


Figure: BTP-II 16-week implementation Gantt chart.

16-Week Deliverables Schedule

- **W1–W3:** Detailed CAD modeling (.STEP assembly & 2D drawings).
- **W4–W6:** Ansys hyperelastic FEA (simulation dataset & report).
- **W7–W8:** FDM TPU 3D printing & Dyneema rigging (hardware prototype).
- **W9–W11:** Analog sensor PCB & ESP32 FreeRTOS firmware.
- **W12–W14:** Static calibration & pull-out force trials.
- **W15–W16:** Live slip-prevention demonstration & defense.

Expected Outcomes & Final Demonstration (BTP-II)

Tangible engineering deliverables to be presented to the evaluation committee

1. Autonomous Grasping

- Conforming gently to diverse shapes: spheres, cylinders, fragile eggs, soft fruits.
- Zero crushing damage on fragile surfaces.
- Payload capacity of 800–1500 g on a gripper self-weight of <450 g (Payload ratio >2:1).

2. Real-Time Telemetry

- Live graphical dashboard streaming contact force and FFT spectral density.
- Integrated micro-ROS client publishing to ROS 2 on host workstation.
- Deterministic multi-tasking executing on ESP32 dual cores.

3. Slip Prevention Demo

- Live physical test: external dislodging tension applied to a grasped object.
- On-chip FFT detects tactile micro-vibrations in 100–300 Hz band.
- Servomotors automatically tighten grip within <35 ms, arresting slip before drop.

Conclusion & Key References

Summary of Stage-I accomplishments and primary citations

Conclusions

- Stage-I establishes a complete, technically sound research direction combining compliant mechanics, additive manufacturing, tactile perception, and real-time edge control.
- Monolithic FDM TPU printing eliminates mold delays and chemical hazards of silicone casting.
- Tendon-driven base servomotors optimize payload-to-weight ratio while maintaining safe contact compliance.
- The integrated FreeRTOS / FFT slip detection framework provides deterministic autonomy without bulky optical cameras.
- The 16-week BTP-II roadmap establishes measurable milestones leading to a validated physical prototype.

Key Verifiable References

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